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Batch no -074

Topic – Linux File permission

- **File permission in linux:**

File permission control who can read, right and execute directory

- **Permission types:**

Permission	Symbol	Meaning
Read	r	View/read file content
Write	w	Modify/edit/append file
Execute	x	Run file directory

- **Permission groups:**

Permission are divided into 9 characters, 9 characters divided in 3 group owner, group, other.

Example: rwx --x -wx.

Group	Meaning
Owner(O)[rwx]	File creator
Group(G)[--x]	Users in same group
Others(O)[-wx]	Everyone else

- **Numeric permissions**

Number	Permission
4	Read (r)

Number	Permission
2	Write (w)
1	Execute (x)

- Files

File type	Meaning
-	Regular file
d	Directory
l	Link and library

To change the permission in linux “chmod_code_filename” command is used.

- **How to read linux command**?

drwxr-xr-x. 3 root root 22 Jan 22 16:31 home

d – directory

rwxr-xr-x. – permission

3 – hard link files

Root – owner

Root – group

22 jan 22 16:31– date, month and time.

```
root@ip-172-31-14-185:/fusion
Connection to ec2-65-1-147-244.ap-south-1.compute.ama
zonaws.com closed.
client_loop: send disconnect: Connection reset by pee
r

HP@shravani MINGW64 ~/downloads
$ ssh -i "demo-key.pem" ec2-user@ec2-65-1-147-244.ap-south-1.compute.amazo
naws.com
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Jan 22 16:32:23 2026 from 106.193.204.28
[ec2-user@ip-172-31-14-185 ~]$
[ec2-user@ip-172-31-14-185 ~]$
[ec2-user@ip-172-31-14-185 ~]$
[ec2-user@ip-172-31-14-185 ~]$ sudo su -
Last login: Thu Jan 22 16:32:49 UTC 2026 on pts/1
[root@ip-172-31-14-185 ~]#
[root@ip-172-31-14-185 ~]# cd /
[root@ip-172-31-14-185 /]#
[root@ip-172-31-14-185 /]#
[root@ip-172-31-14-185 /]# mkdir fusion
[root@ip-172-31-14-185 /]#
[root@ip-172-31-14-185 /]#
[root@ip-172-31-14-185 /]# cd fusion/
[root@ip-172-31-14-185 fusion]# ls
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# vi aws.txt
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# ls -l
total 4
-rw-r--r--. 1 root root 35 Jan 22 16:59 aws.txt
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# ls -l
total 4
-rw-r--r--. 1 root root 35 Jan 22 16:59 aws.txt
[root@ip-172-31-14-185 fusion]# |
```

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total 4
-rw-r--r--. 1 root root 35 Jan 22 16:59 aws.txt
[root@ip-172-31-14-185 fusion]# cat aws.txt
hello i'm fusion nice to meet you
[root@ip-172-31-14-185 fusion]# ls -l
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# 2 16:59 aws.txt
[root@ip-172-31-14-185 fusion]# ls
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# ls-ll
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# ls -ll
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# 2 16:59 aws.txt
[root@ip-172-31-14-185 fusion]# chmod 444 aws.txt
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# vi aws.txt
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# ls -l
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-r--r--r--. 1 root root 35 Jan 22 16:59 aws.txt
[root@ip-172-31-14-185 fusion]#
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[root@ip-172-31-14-185 fusion]#
[root@ip-172-31-14-185 fusion]# ls -l
total 4
-r--r--r--. 1 root root 35 Jan 22 16:59 aws.txt
[root@ip-172-31-14-185 fusion]# history
    1  cd /
    2  mkdir fusion
    3  cd fusion/
    4  ls
    5  vi aws.txt
    6  ls -l
    7  cat aws.txt
    8  ls -l
    9  ls
   10  ls-ll
   11  ls -ll
   12  chmod 444 aws.txt
   13  vi aws.txt
   14  ls -l
   15  history
[root@ip-172-31-14-185 fusion]# |
```

```
root@ip-172-31-14-185:/fusion
hello i'm fusion nice to meet you
~
~
~
```

```
root@ip-172-31-14-185:/
total 32
lrwxrwxrwx.   1 root root     7 Jan 30  2023 bin -> usr/bin
dr-xr-xr-x.   5 root root 16384 Jan  2 00:01 boot
drwxr-xr-x.  15 root root  3040 Jan 22 16:31 dev
drwxr-xr-x.  76 root root 16384 Jan 22 16:31 etc
drwxr-xr-x.   2 root root    21 Jan 22 17:29 fusion
drwxr-xr-x.   3 root root    22 Jan 22 16:31 home
lrwxrwxrwx.   1 root root     7 Jan 30  2023 lib -> usr/lib
lrwxrwxrwx.   1 root root     9 Jan 30  2023 lib64 -> usr/lib64
drwxr-xr-x.   2 root root     6 Jan  1 23:58 local
drwxr-xr-x.   2 root root     6 Jan 30  2023 media
drwxr-xr-x.   2 root root     6 Jan 30  2023 mnt
drwxr-xr-x.   3 root root    17 Jan  2 00:01 opt
dr-xr-xr-x. 169 root root     0 Jan 22 16:31 proc
dr-xr-x---.   3 root root   119 Jan 22 17:30 root
drwxr-xr-x.  27 root root   820 Jan 22 16:31 run
lrwxrwxrwx.   1 root root     8 Jan 30  2023 sbin -> usr/sbin
drwxr-xr-x.   2 root root     6 Jan 30  2023 srv
dr-xr-xr-x.  13 root root     0 Jan 22 16:31 sys
drwxrwxrwt.  11 root root   220 Jan 22 17:30 tmp
drwxr-xr-x.  12 root root   144 Jan  1 23:59 usr
drwxr-xr-x.  19 root root   266 Jan 22 16:31 var
[root@ip-172-31-14-185 /]# chmod 055 fusion
[root@ip-172-31-14-185 /]#
[root@ip-172-31-14-185 /]# ls -l
total 32
lrwxrwxrwx.   1 root root     7 Jan 30  2023 bin -> usr/bin
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drwxr-xr-x.  15 root root  3040 Jan 22 16:31 dev
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